+33 6 29 73 70 51
ugo.nzongani@lis-lab.fr

Ugo Nzongani

Education

PhD Candidate in Quantum Computing **2023 - 2026 (Ongoing)**
Aix-Marseille Université, ENSTA, Institut Polytechnique de Paris

Subject: Engineering quantum simulations and optimization algorithms with quantum walks
Supervisors: Prof. Giuseppe Di Molfetta, Prof. Andrea Simonetto

Master of Quantum and Distributed Computer Science with honours **2021 - 2023**
Paris-Saclay Université

Main courses: Quantum Computing, Quantum Information, High Performance Computing, Distributed Algorithms, Advanced Algorithmic, Optimization, Complexity Theory

Bachelor of Computer Science with honours **2018 - 2021**
Paris-Saclay Université

Main courses: Functional Programming, Data Structures, Algorithmic, Machine Learning, C/Python/Java/Assembly Programming, Logic

Baccalauréat Scientifique, equivalent to High School GB A levels with honours **2018**
Fustel de Coulanges high school, Massy

Main courses: Mathematics, Physics, Biology

Work Experience

Volunteering - LOC **April 2023**
Ecole Polytechnique - International Physicists' Tournament *Palaiseau - France*

- I participated in the 15th edition of the IPT as a member of the Local Organizing Committee. This is an international tournament for physics students where so-called "Physics Fight" takes place four days of the week. A fight is a scientific debate between 3 teams around challenge they have been studying for several months.

Research Internship **5 months - 2023**
Paris-Saclay Université - Laboratoire de Méthodes Formelles - QuaCS Team *Orsay - France*

- Subject: Dirac quantum walk on tetrahedra
  Supervisor: Prof. Pablo Arrighi

Math tutor **7 months - 2023**
Acadomia *Massy - France*

- Private math lessons to high school students

Research Internship **2 months - 2022**
INRIA - Laboratoire de Méthodes Formelles - QuaCS Team *Orsay - France*

- Subject: Quantum circuits for quantum walk with position-dependent coin operators
  Supervisor: Dr. Pablo Arnault

Distinctions

- Quantum-Saclay Scholarship, March 2023

Publications

- Nzongani, U., Laplace Mermoud, D., G, Di Molfetta, Simonetto, A.Sampled-Based Guided Quantum Walk: Non-variational quantum algorithm for combinatorial optimization. Quantum 10, 2234 (2026).
- Nzongani, U., Laplace Mermoud, D., A, Braida, A.Scaling QAOA: transferring optimal adiabatic schedules from small-scale to large-scale variational circuits. IEEE, International Conference on Quantum Control, Computing and Learning (2026)
- Nzongani, U., Simonetto, A., Di Molfetta, G.Non-unitary enhanced transfer efficiency in quantum walk search on complex networks. Physical Review A 112, 052451 (2025).
- Zylberman, J., Nzongani, U., Simonetto, A., Debbasch, F.Efficient Quantum Circuits for Non-Unitary and Unitary Diagonal Operators with Space-Time Accuracy trade-offs. ACM Transaction on Quantum Computing (2025).
- Nzongani, U., Eon, N., Márquez-Martín, I. et al. Dirac quantum walk on tetrahedra. Physical Review A 110, 042418 (2024).
- Nzongani, U., Arnault P.Adjustable-depth quantum circuit for position-dependent coin operators of discrete-time quantum walks. Quantum Information Processing 23, 193 (2024).
- Nzongani, U., Zylberman, J., Doncecchi, CE. et al. Quantum circuits for discrete-time quantum walks with position-dependent coin operator. Quantum Information Processing 22, 270 (2023).

Preprints

- Nzongani, U., Laplace Mermoud, D., Braida, A.Gap-Transfer QAOA.

Talks

- Exploring quantum approaches to combinatorial optimization, *CERFACS, Toulouse*
- Scaling QAOA: transferring optimal adiabatic schedules from small-scale to large-scale variational circuits, *2026 IEEE International Conference on Quantum Control, Computing and Learning, Aalborg, Denmark*
- Sampled-Based Guided Quantum Walk: Non variational quantum algorithm for combinatorial optimization, *Seminar LaBRI, Bordeaux*
- Sampled-Based Guided Quantum Walk: Non variational quantum algorithm for combinatorial optimization, *Quantum days AMU 2025, CIRM, Marseille*
- Sampled-Based Guided Quantum Walk: Non variational quantum algorithm for combinatorial optimization, *PGMO DAYS 2025, EDF LAB, Saclay*
- Hybrid Quantum Search on Complex Networks: Noisy Spatial Search with Quantum Stochastic Walks, *PGMO DAYS 2024, EDF LAB, Saclay*
- Hybrid Quantum Search on Complex Networks: Noisy Spatial Search with Quantum Stochastic Walks, *Quantum ETS Seminar 2024, Montréal*

Summer Schools

June 7th - 19th 2026
Towards Fault Tolerant Quantum Computing, AlgoLab, Institut quantique de Sherbrooke, Bishop's University - Sherbrooke, Canada

May 23th - 28th 2025
Mathematics and Physics of Quantum Computing and Quantum Learning, Aix-Marseille Université, Université de Toulouse - Porquerolles, France
June 2nd - 14th 2024
Near-Term Quantum Algorithms AlgoLab, Institut quantique de Sherbrooke - Centre de Villégiature Jouvence, Canada

Skills

Programming	Python, OCaml, Java, C, C++, Julia, LaTeX, HTML, CSS, PHP
Communication	French (native), English (885/990 TOEIC)